

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES							
10		VB2					***** BALL VALVES *****			810	NIPPLE TO BE CUT FROM SCH 160 PIPE.						
20										B1	ALL BOLTS ARE METRIC DIA.						
20	U04		BALL	A01AC01	3	3	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB LO	0	0	B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1						
30	U04		BALL	A01AC02	3	3	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL RB LO	0	0		INCH 1500# PERAR BALL VALVES ITEM CODE						
40	U04		BALL	B01AC01	0.75	2	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0		B15AM10						
50	U04	VA5	BALL	B01AC02	3	3	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO	0	0	FA8	USE THREADED FLANGES FOR INSTRUMENT						
60	U04	VA5	BALL	B01AC07	6	8	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0		CONNECTIONS ONLY.						
70							***** GATE VALVES *****			VA5	USE THESE VALVES ONLY WHERE SPECIFIED						
70											ON PIPING AND INSTRUMENT DIAGRAMS						
80										VB2	SOFT SEATED BALL VALVES ARE LIMITED TO						
80	U04		GATE	G01AC48	0.75	1.5	GATE 150# RF LTCS A350 LF2 CL.1 TRIM 16 WEDGE HW	0	0		MAXIMUM TEMPERATURE OF 200°C.						
90			GATE	G01AC41	2	3	GATE 150# RF LTCS A352 LCC TRIM 16 WEDGE HW	0	0								
100			GATE	W01AC70	3	3	GATE 150# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW	0	0								
110							***** PIPE ABOVE GROUND *****										
110																	
120																	
120	U04		PIPE	5358508	0.5	1.5	PIPE SCH XS SMLS LTCS A333 GR.6 PE	0	0								
130			PIPE	5358514	2	2	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0								
140			PIPE	5358505	3	3	PIPE SCH STD SMLS LTCS A333 GR.6 BE	0	0								
150		810	PIPE	5358512	0.5	0.5	PIPE SCH 160 SMLS LTCS A333 GR.6 PE	0	0								
160							***** PIPE BELOW GROUND *****										
160																	
170																	
170			PIPE	9001254CTD	6	8	LINEPIPE SMLS WT= 6.35 MM API-5L X60 BE COATED	0	0								
180							***** FITTINGS ABOVE GROUND*****										
180																	
190																	
190			ELBOW	5392600	0.75	2	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6	0	0								
200			ELBOW	5392438	3	3	ELL BW 90 DEG LR SCH STD LTCS A420 WPL6	0	0								
210			ELBOW	5392606	0.75	2	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6	0	0								
220			ELBOW	5392444	3	3	ELL BW 45 DEG LR SCH STD LTCS A420 WPL6	0	0								
230																	
230			TEE	5392609	0.75	2	TEE BW SCH XS LTCS A420 WPL6	0	0								
240			TEE	5392447	3	3	TEE BW SCH STD LTCS A420 WPL6	0	0								
250			TEE RED	5392661	0.75	1.5	TEE RED BW SCH XS LTCS A420 WPL6	1	2								
260			TEE RED	5392454	1.5	2	TEE RED BW SCH STD/XS LTCS A420 WPL6	3	3								
270																	
270			CAP	5392610	0.75	2	CAP BW SCH XS LTCS A420 WPL6	0	0								
280			CAP	5392448	3	3	CAP BW SCH STD LTCS A420 WPL6	0	0								
290	U04		CAP	9005055	6	8	CAP BW WT 6.35MM LTCS A860 WPHY-60	0	0								
300																	
300			RED CONC	5392458	0.75	1.5	REDUCER CONC SCH XS LTCS A420 WPL6	1	2								
310			RED CONC	5392452	1	2	REDUCER CONC SCH STD/XS LTCS A420 WPL6	3	3								
320			RED ECC	5392461	0.75	1.5	REDUCER ECC SCH XS LTCS A420 WPL6	1	2								
330			RED ECC	5392453	1	2	REDUCER ECC SCH STD/XS LTCS A420 WPL6	3	3								
340							*****INDUCTION BENDS ABOVE GROUND*****										
340																	
350																	
350			INDUC BEND	9001813	6	8	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0								
360			INDUC BEND	9001823	6	8	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0								
370			INDUC BEND	9001833	6	8	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW	0	0								
380							*****INDUCTION BENDS BELOW GROUND*****										
380																	
390																	
390			INDUC BEND	9001813CTD	6	8	INDUCTION BEND 90 DEG X 5D RADIUS WT 6.35MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0								
400			INDUC BEND	9001823CTD	6	8	INDUCTION BEND 45 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0								
410			INDUC BEND	9001833CTD	6	8	INDUCTION BEND 22.5 DEG X 5D RADIUS WT 6.35 MM API 5L X60 C/W 1.5D TANGENT ENDS BW COATED	0	0								
420																	
420	U04		OLETS	5505335	0.75	1	WELDOLET SCH XS LTCS A350 LF2 CL.1	2	2								
430	U04		OLETS	5505179	0.75	2	WELDOLET SCH STD/XS LTCS A350 LF2 CL.1	3	3								
440	U04		OLETS	9006407	0.75	2	WELDOLET WT 6.35 MM/XS WT LTCS A694 F60	6	8								
450																	
450	U04		OLETS	5504001	0.75	1.5	NIPOFLANGE 150# RF SCH XS LTCS A350 LF2 CL.1	2	2								
460	U04		OLETS	5504002	0.75	1.5	NIPOFLANGE 150# RF SCH STD/ XS LTCS A350 LF2 CL.1	3	3								
470	U04		OLETS	9006801	0.75	2	NIPOFLANGE 150# RF 6.35 MM/XS WT LTCS A694 F60	6	8								
480	U04		OLETS	5504020	0.75	1.5	NIPOFLANGE 300# RF SCH XS LTCS A350 LF2 CL.1	2	2								
490	U04		OLETS	5504021	0.75	1.5	NIPOFLANGE 300# RF SCH STD/XS LTCS A350 LF2 CL.1	3	3								
500							***** FLANGES ABOVE & BELOW GROUND*****										
500																	
510																	
510	U04	FA8	TEE RED	5599692	0.5	0.5	FLG SCRDR (FNPT) REDUCING 150# RF LTCS A350 LF2 CL.1	0.75	1.5								
520	U04	FA8	TEE RED	5599703	0.5	0.5	FLG SCRDR (FNPT) REDUCING 300# RF LTCS A350 LF2 CL.1	0.75	2								
			TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN		CONSTRUCTION		CORR.ALLOW.		SERVICE		MATERIAL		DATE	BY	REV	SHEET	PIPING LINE CLASS
					ALL SIZES - BUTT WELD SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE		CORR. ALLOW. 1.5 MM		SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 FLUID CODES - PIM-SU-5112-TCO		MATERIAL:LTCS FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 75 °C		12/16/2024	BY OZAC	U04	Page 1 of 2	150H5E

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530										
530	U04		FLG ADAPTER	5851200	0.75	2	FLANGE ADAPTER 150# RF 316 SS X 12 MM TUBE	0	0	
540										
540	U04		FLANGE	5556264	0.5	2	FLG WN 150# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
550	U04		FLANGE	5556261	0.75	3	FLG WN 150# RF LTCS A350 LF2 CL.1 SCH STD	0	0	
560	U04		FLANGE	5557033	3	3	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH STD	0	0	
570	U04		FLANGE	5557036	2	2	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
580	U04		FLANGE	5555935	3	3	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH STD	0	0	
590	U04		FLANGE	5557393	0.75	2	FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
600			FLANGE	9001473	6	8	FLG WN 150# RF LTCS A694 F60 WT 6.35MM	0	0	
610	U04		BLIND	5615566	0.5	3	FLG BLIND 150# RF LTCS A350 LF2 CL.1	0	0	
620			BLIND	9001449	6	8	FLG BLIND 150# RF LTCS A694 F60	0	0	
630	U04		BLIND	5615616	0.75	2	FLG BLIND 1500# RF LTCS A350 LF2 CL.1	0	0	
640										
640							***** GASKETS *****			
650										
650			GASKET	5673043	0.75	8	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
660	U04		GASKET	5673103	2	3	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
670			GASKET	5672251	0.75	2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
680										
680	B1						***** BOLTING *****			
690										
690	U04		BOLTING	5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
700										
700			SPEC BLIND	5749580	0.75	10	SPECTACLE BLIND 150# RF LTCS A 516 GR.70	0	0	
710										
710							***** PIPE NIPPLES *****			
720										
720	810			0			NIPPLE PBE CUT FROM PIPE			
730										

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD SEAL WELDING - REQUIRED PIPE BENDS - INDUCTION BENDS THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1.5 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.8 ABOVE GROUND FACTOR = 0.5 BELOW GROUND FACTOR = 0.6 FLUID CODES - PIM-SU-5112-TCO	MATERIAL:LTCS FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 75 °C	12/16/2024	BY OZAC	U04	Page 2 of 2	150H5E